

ECE264D:

CMOS Analog Integrated Circuits and Systems IV

Lecture 12:

All Digital Phase Locked Loops

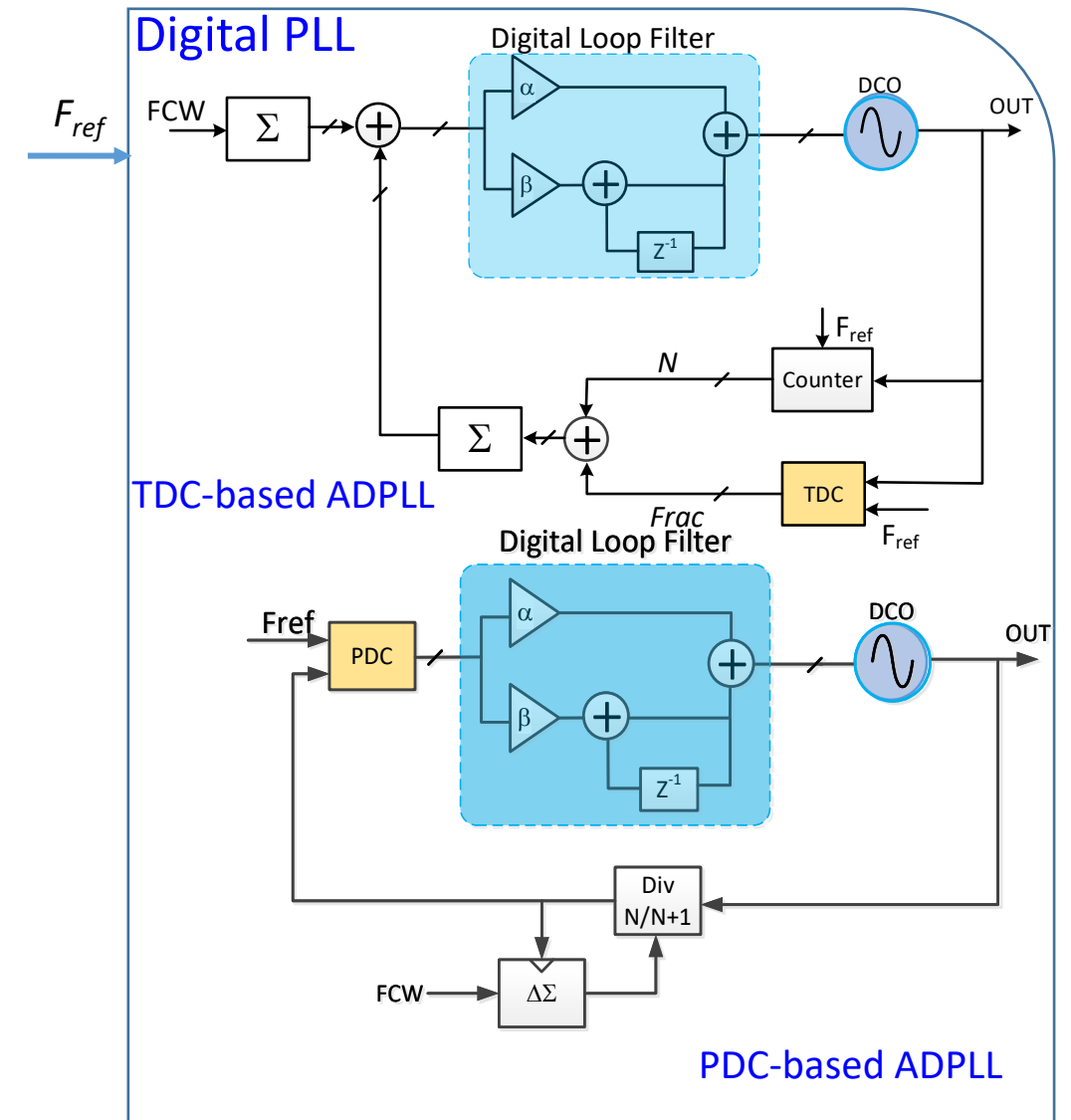
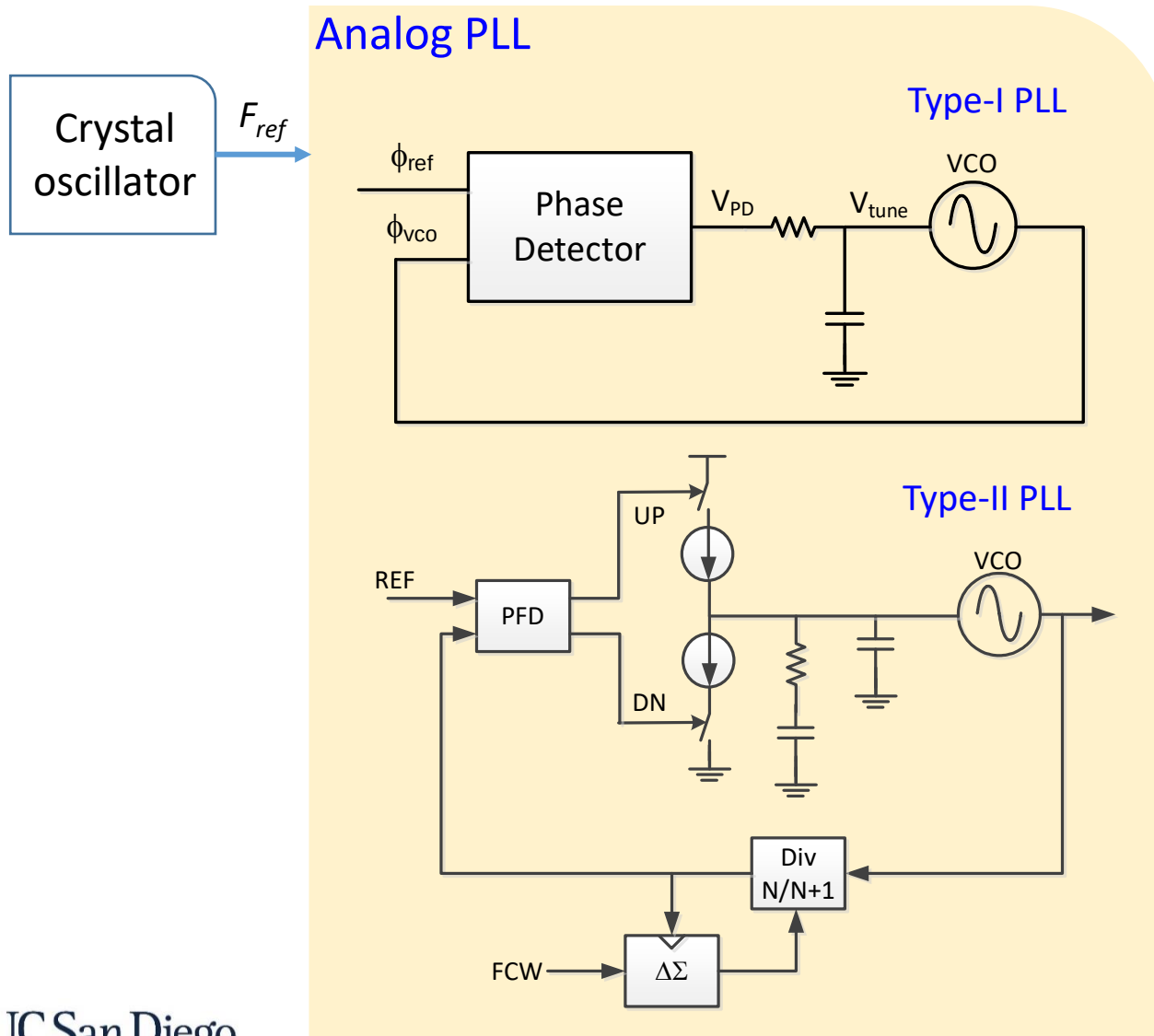
Digital Loop Filter

Digitally Controlled Oscillator

Yiwu Tang



Frequency Synthesizer Overview

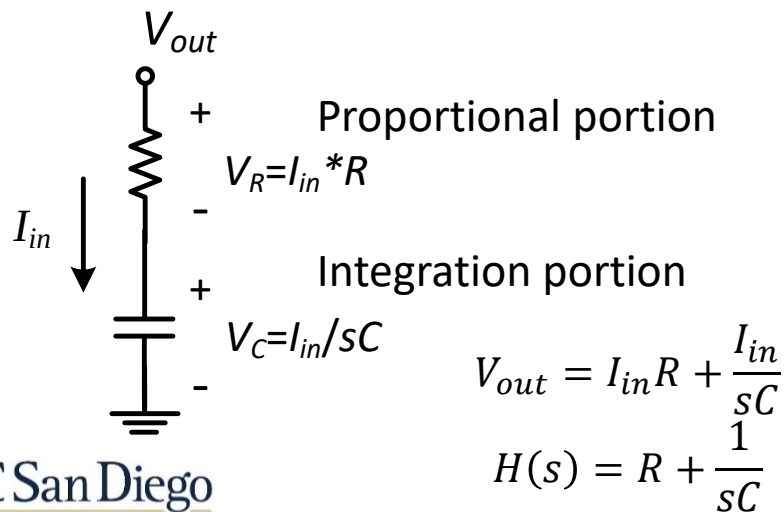
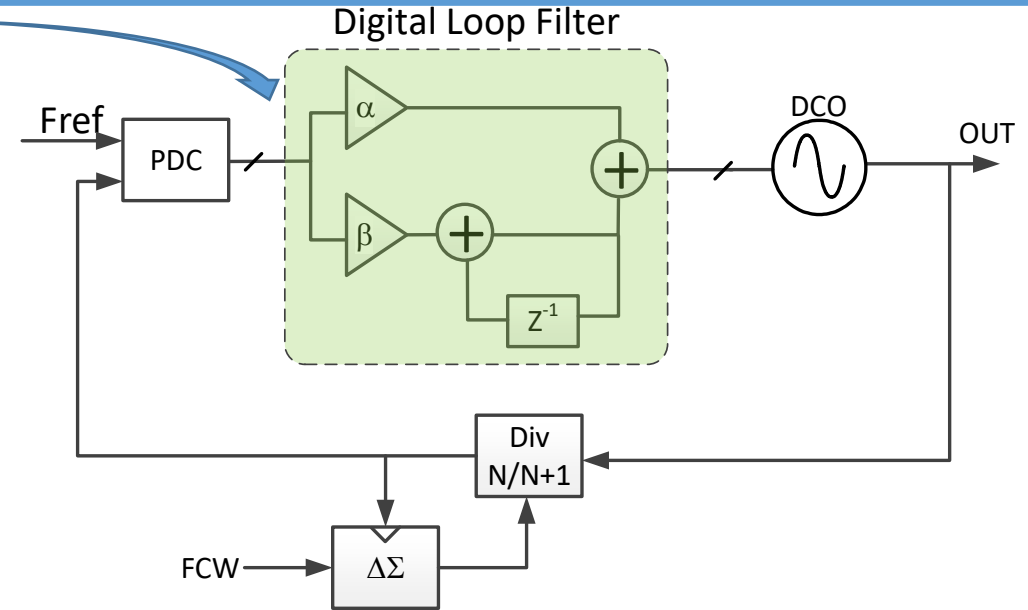
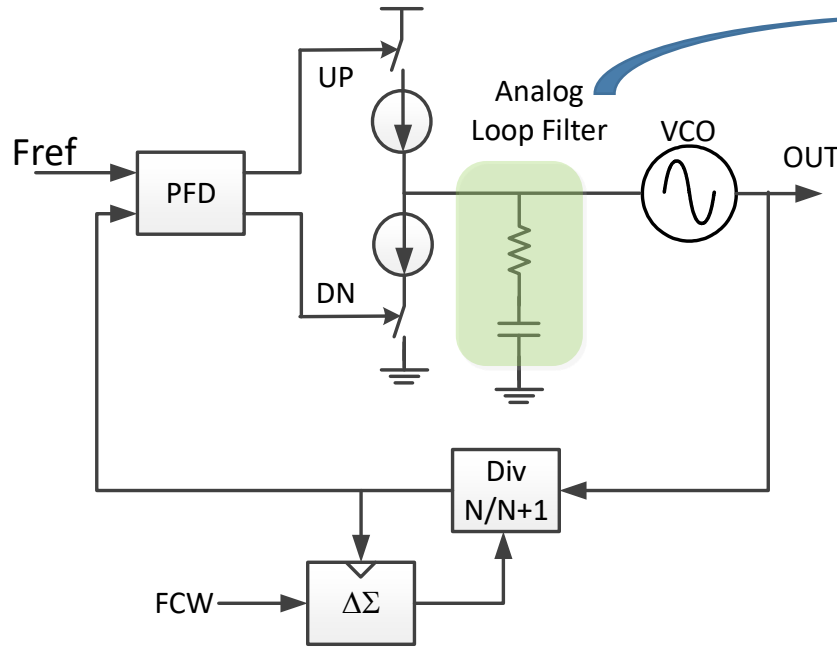


Outline

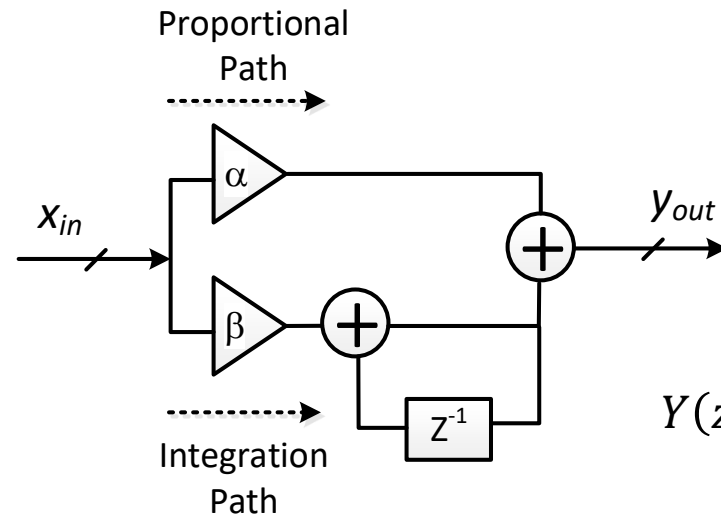
- Digital signal process basics review
 - Laplace transform and s-domain transfer function
 - Z-transform and z-domain transfer function
 - Bilinear transform from s-domain to z-domain transfer function
- ADPLL building block – Digital loop filter
 - Design digital filters based on their analog prototypes
 - Simple RC filter
 - Higher order filters
- Digitally controlled oscillator (DCO)
 - Digital control
 - Resolution and nonlinearity



Digital Loop Filter Design



$R, C \Rightarrow \alpha, \beta$

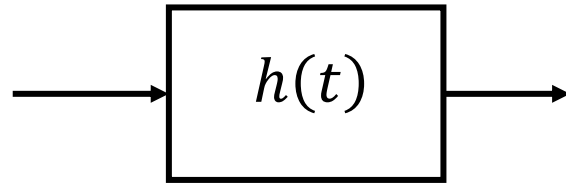


$$Y(z) = \left(\alpha + \frac{\beta}{1 - z^{-1}} \right) X(z)$$



Signal Processing Basics Review

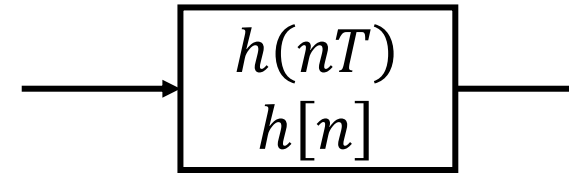
Continuous Time



- Impulse response $h(t)$
- s-domain transfer function with Laplace transform

$$H(s) = \int_0^{\infty} h(t)e^{-st} dt$$

Discrete Time



- Impulse response $h[n]$
- z-domain transfer function with Z transform

$$H(z) = \sum_{n=0}^{\infty} h[n]z^{-n}$$

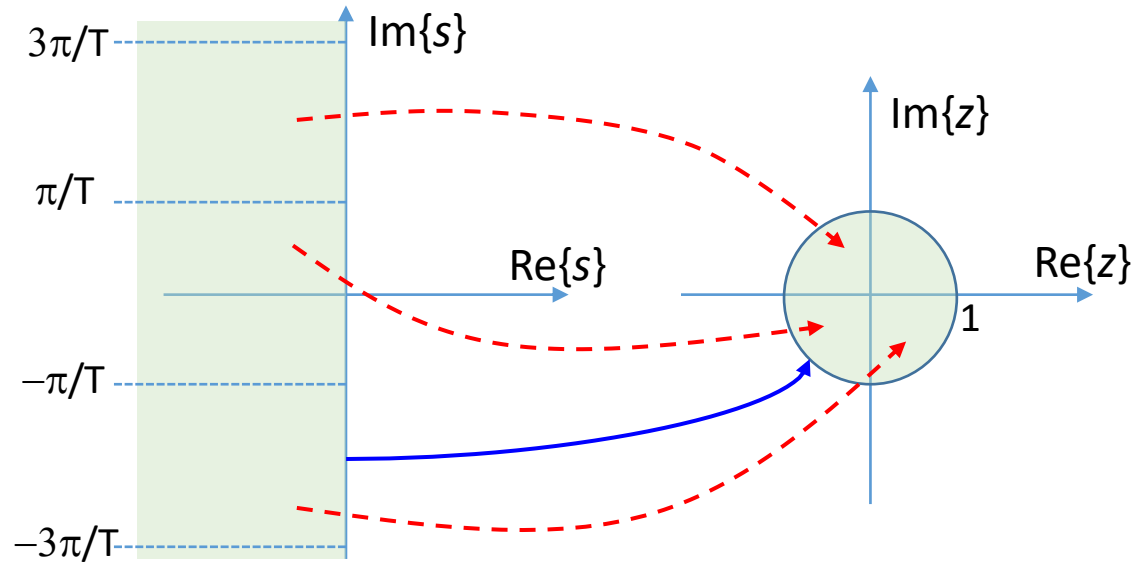


$$e^{sT} = z$$

Impulse invariance mapping



Signal Processing Basics Review

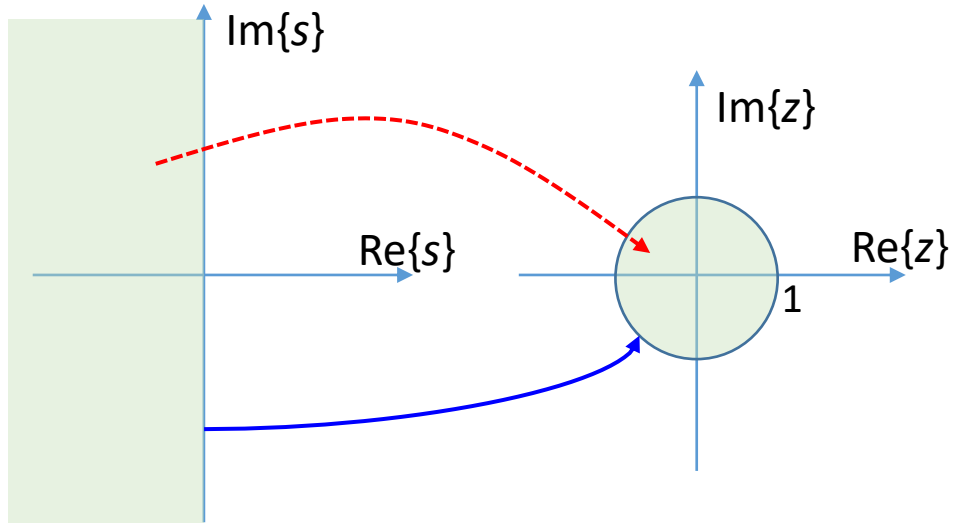


S domain	Z domain
Left-hand plane	Inside unit circle
Imaginary axis	Unit circle
Right-hand plane	Outside unit circle

- $z = re^{j\omega_d}$, $s = \sigma + j\omega_c$
 - The discrete-time frequency axis
 $\omega_d \in (-\pi/T, \pi/T)$
 - The continuous-time frequency axis
 $\omega_c \in (-\infty, \infty)$
- **Impulse invariance transform $z = e^{sT}$**
 - $re^{j\omega_d} = e^{\sigma T} e^{j\omega_c T}$
 $\Rightarrow r = e^{\sigma T}$ and $\omega_d = \omega_c T$
 - $\sigma = 0, r = 1$; $\sigma < 0, r < 1$; $\sigma > 0, r > 1$
 - For $\omega_c \in (-\infty, \infty)$, s and $s + k(j 2\pi/T)$ map to the same $z \Rightarrow$ aliasing
 - Impulse invariance mapping applicable only to bandlimited signal where $-\pi/T < \omega_c < \pi/T$



Signal Processing Basics Review



S domain	Z domain
Left-hand plane	Inside unit circle
Imaginary axis	Unit circle
Right-hand plane	Outside unit circle

- Bilinear Transform to avoid aliasing

$$s = \frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}, \quad z = \frac{1+(T_s/2)s}{1-(T_s/2)s}$$

- $z = re^{j\omega_d}, s = \sigma + j\omega_c$

For $\sigma=0$, $z = \frac{1+j\omega_c(T_s/2)}{1-j\omega_c(T_s/2)}$

$\Rightarrow |z| = 1$, imaginary axis maps to unit circle.

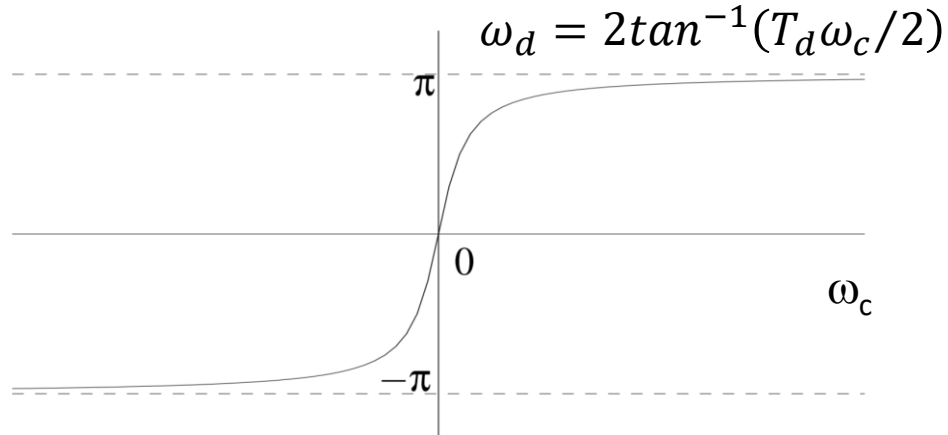
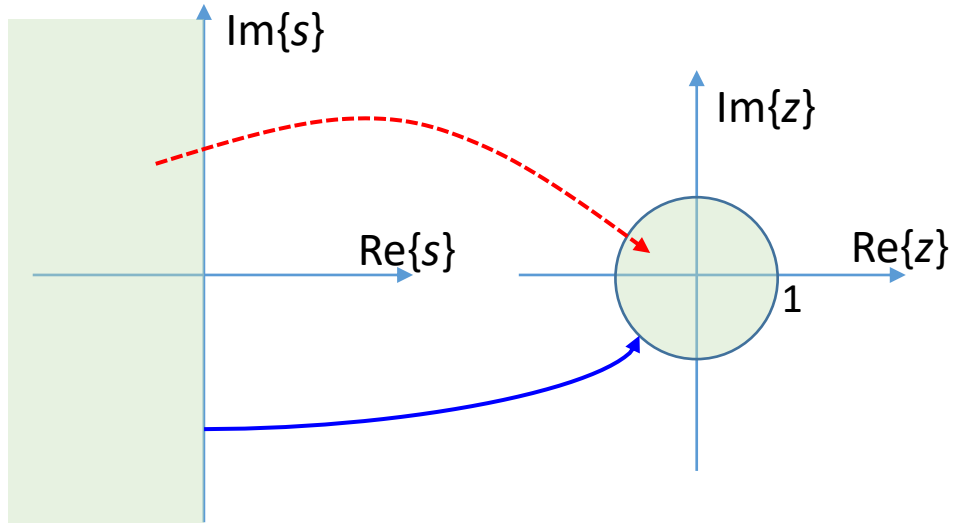
For $\sigma \neq 0$, $z = \frac{\left(1+\frac{T_s}{2}\sigma\right)+j\omega_c(T_s/2)}{\left(1+\frac{T_s}{2}\sigma\right)-j\omega_c(T_s/2)}$

$\Rightarrow \sigma < 0, r < 1$: left-hand plane maps to inside of unit cycle;

$\Rightarrow \sigma > 0, r > 1$: right-hand plane maps to outside of unit cycle;



Signal Processing Basics Review



$\omega_d \leftrightarrow \omega_c$ has a 1:1 mapping $\Rightarrow$ No aliasing

- Bilinear Transform to avoid aliasing

$$s = \frac{2}{T_s} \frac{1-z^{-1}}{1+z^{-1}}, \quad z = \frac{1+(T_s/2)s}{1-(T_s/2)s}$$

$$\sigma + j\omega_c = \frac{2}{T_s} \frac{1 - re^{-j\omega_d}}{1 + re^{-j\omega_d}}$$

For $r=1$,

$$\sigma + j\omega_c = \frac{2}{T_s} \frac{(e^{j\omega_d/2} - e^{-j\omega_d/2})e^{-j\omega_d/2}}{(e^{j\omega_d/2} + e^{-j\omega_d/2})e^{-j\omega_d/2}} = j \frac{2}{T_s} \tan\left(\frac{\omega_d}{2}\right)$$

$$\Rightarrow \omega_c = \frac{2}{T_s} \tan\left(\frac{\omega_d}{2}\right), \quad \omega_d = 2 \tan^{-1}(T_s \omega_c / 2)$$

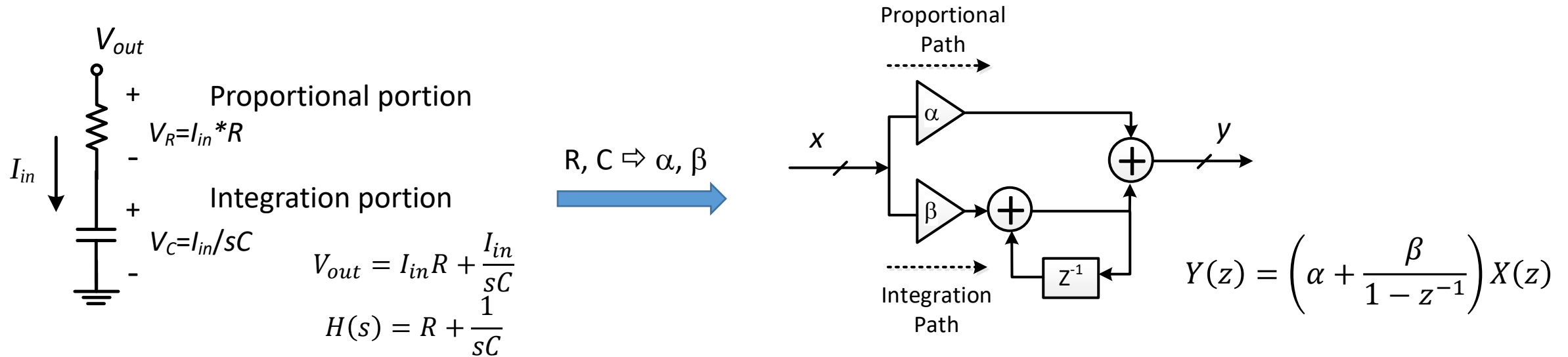
Euler's formula:

$$\cos x = \operatorname{Re}(e^{ix}) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\sin x = \operatorname{Im}(e^{ix}) = \frac{e^{ix} - e^{-ix}}{2i}$$



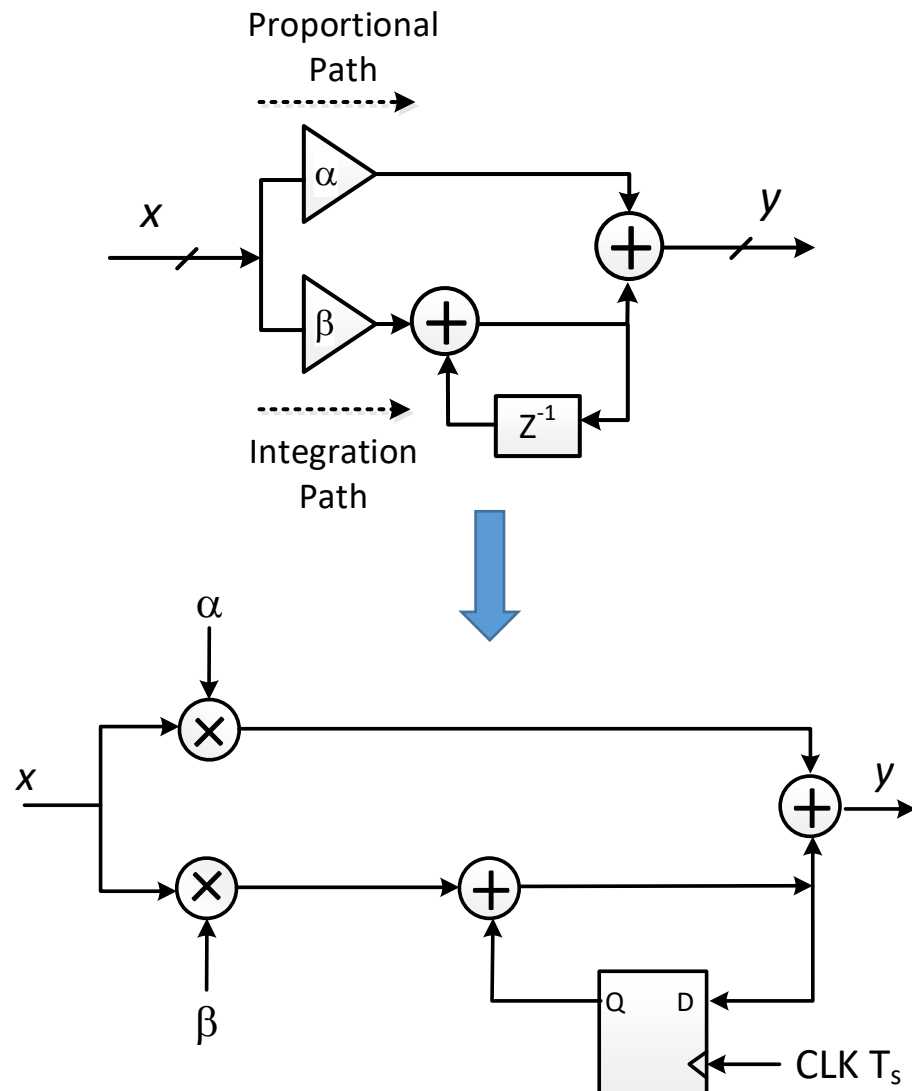
Transform from an Analog to a Digital Loop Filter



- $H(s) = R + \frac{1}{sC}, s = \frac{2}{T_s} \frac{1 - z^{-1}}{1 + z^{-1}} \Rightarrow H(z) = \frac{\left(\frac{T_s}{2C} + R \right) + z^{-1} \left(\frac{T_s}{2C} - R \right)}{1 - z^{-1}}$
- On the other hand, $H(z) = \frac{(\alpha + \beta) + z^{-1}(-\alpha)}{1 - z^{-1}}$
- $\Rightarrow \alpha = R - \frac{T_s}{2C}, \beta = \frac{T_s}{C}.$



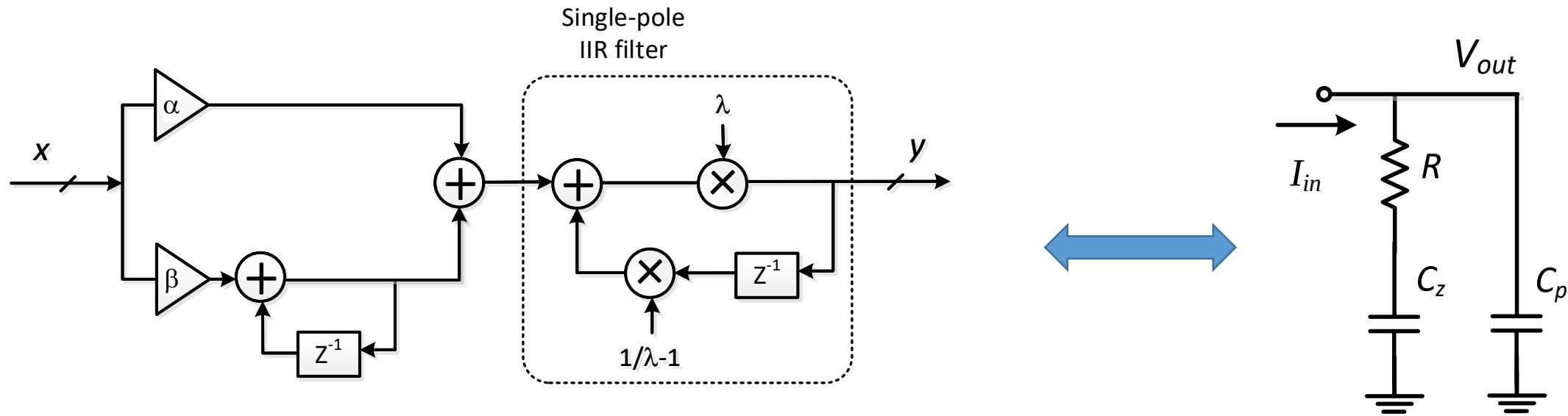
Digital Implementation



- Gain coefficients α and β
 - Approximated as power of 2 (e.g., 2^{-3} , 2^{-5})
 - Multiplication becomes bit shifting
- DFF to implement delay Z^{-1}
- Sufficient bit width in data path for accuracy
- Sign and overflow handling



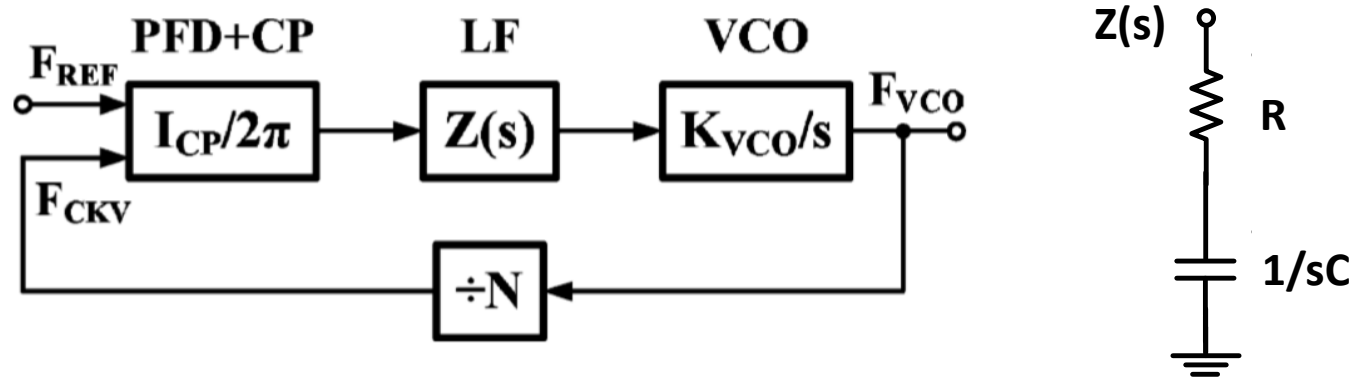
Higher Order Loop Filter



- $\left[(Yz^{-1}) \left(\frac{1}{\lambda} - 1 \right) + W \right] \lambda = Y, \frac{Y}{W} = \frac{\lambda}{1 - (1 - \lambda)z^{-1}}$
- Use bilinear transform, $z = \frac{1 + (T_s/2)s}{1 - (T_s/2)s}$
- For $\lambda \ll 1$ and $f_s = 1/T_s$: $\frac{Y}{W} = \frac{\lambda}{1 - (1 - \lambda) \frac{1 + (T_s/2)s}{1 - (T_s/2)s}} = \frac{1 + \frac{1}{2f_s}s}{1 + \left(\frac{1}{\lambda f_s} - \frac{1}{2f_s} \right)s} \approx \frac{1}{1 + \frac{1}{\lambda f_s}s}$



Design Example

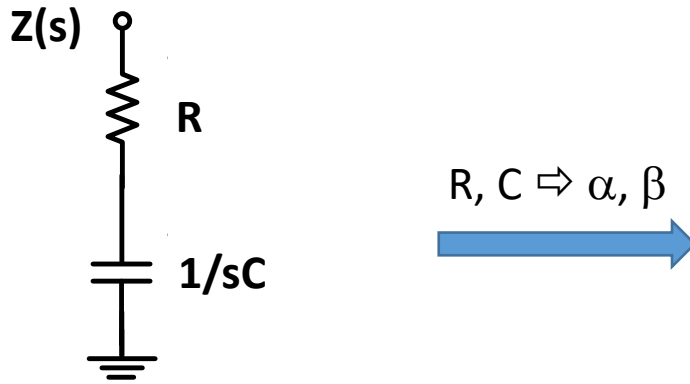


- $H_{OL}(s) = \frac{I_{CP}}{2\pi} \frac{K_V}{s} \frac{1}{N} \left(R + \frac{1}{sC} \right) = \frac{I_{CP}}{2\pi} \frac{K_V}{s} \frac{1}{N} \cdot R \frac{s + \omega_z}{s}$, where $\omega_z = \frac{1}{RC}$
- $\angle H_{OL}(j\omega) = -180 + \arctan\left(\frac{\omega}{\omega_z}\right)$
- $PM = \arctan\left(\frac{\omega_u}{\omega_z}\right)$, where ω_u is unity gain frequency $|H_{OL}(s)| = 1$
- $\omega_z = \frac{\omega_u}{\tan(PM)}$, $R = \frac{\tan(PM)}{C \cdot \omega_u}$
- If $PM=45^\circ$, $\omega_z = \omega_u$

Ref [2]

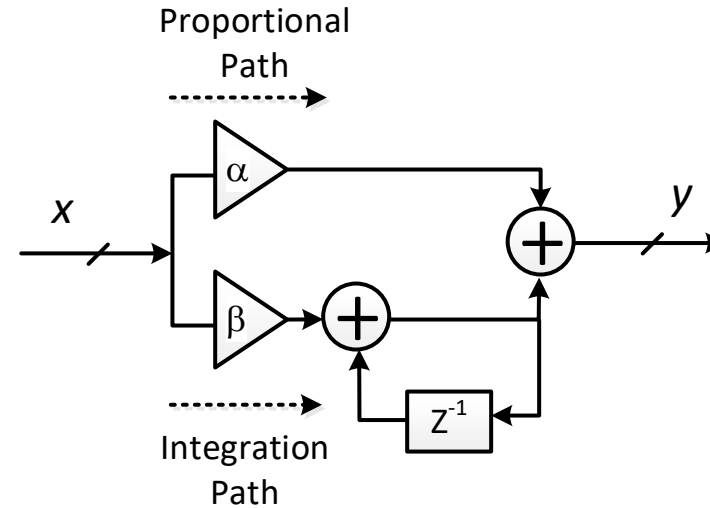


Design Example



- Design requirements/constraints
 - Reference frequency $f_s = 80\text{MHz}$
 - Unity gain frequency $f_u = 1\text{MHz}$
 - Phase margin $\text{PM} = 45^\circ$
- Design parameters
 - $f_u = 1\text{MHz}$, $\text{PM} = 45^\circ \rightarrow f_z = 1\text{MHz}$
 - If $C = 1.4\mu\text{F}$, $f_z = 1\text{MHz} \rightarrow R = \frac{1}{2\pi f_z C} \approx 0.12\Omega$
 - $\alpha = R - \frac{T_s}{2C}$, $\beta = \frac{T_s}{C}$
 - $\alpha = 0.11 \approx 2^{-3}$, $\beta = 0.0089 \approx 2^{-7}$

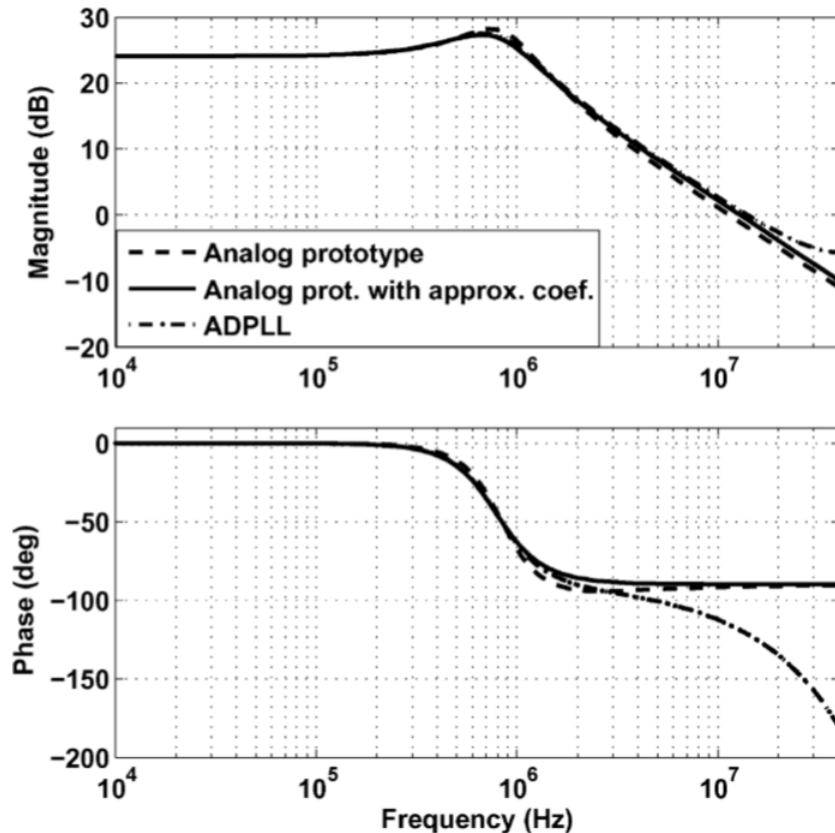
Ref [2]



- $\alpha = R - \frac{T_s}{2C}$, $\beta = \frac{T_s}{C}$
- $\frac{\alpha}{\beta} = \frac{RC}{T_s} - \frac{1}{2} = \frac{\tan(\text{PM})}{T_s \cdot \omega_u} - \frac{1}{2}$
- $\tan(\text{PM}) = T_s \cdot \omega_u \left(\frac{\alpha}{\beta} + \frac{1}{2} \right)$
- Larger α/β ratio yields higher phase margin
- $\alpha/\beta=16$, $\text{PM} \approx 52^\circ$
- $\alpha/\beta=8$, $\text{PM} \approx 33^\circ$
- $\alpha/\beta=32$, $\text{PM} \approx 69^\circ$



Design Example

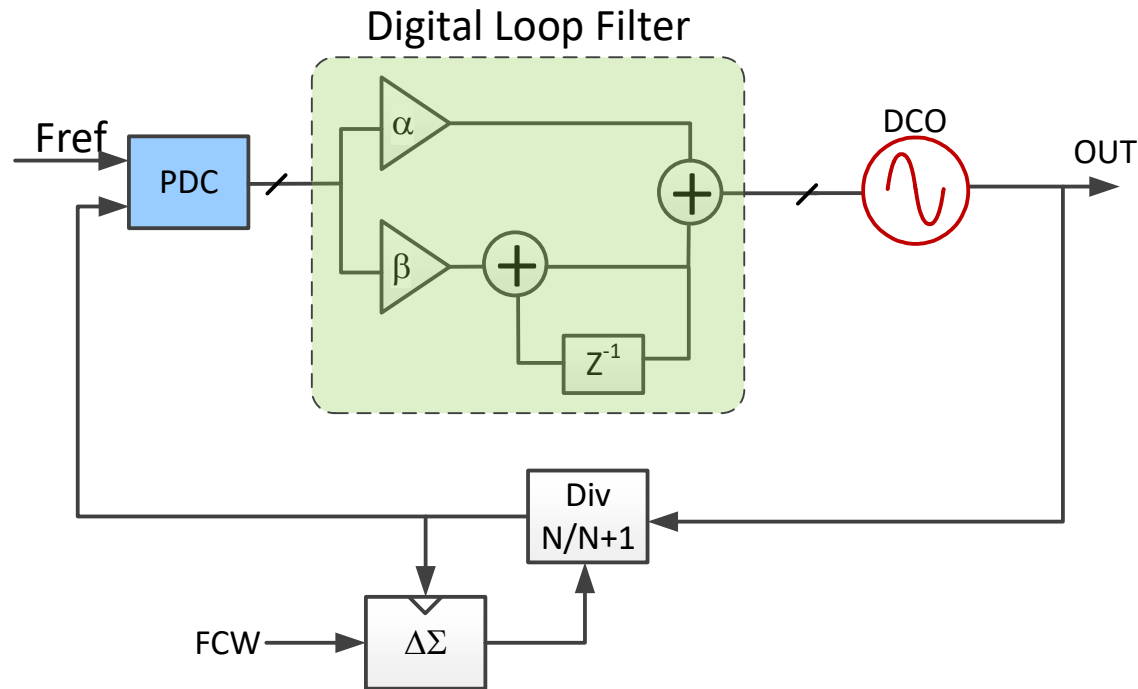


Ref [2]

- Analog prototype and analog prototype with coefficient approximation have very close frequency response
- Digital PLL implementation frequency response is very close to that of analog PLL at low frequency and deviates at frequency close to $f_s/2$
- Deviation at higher frequency because s-domain model is inaccurate when continuous time assumption is no longer valid.
- Delay induced phase margin degradation is clearly shown
- $f_s/f_u \gg 1$ is required

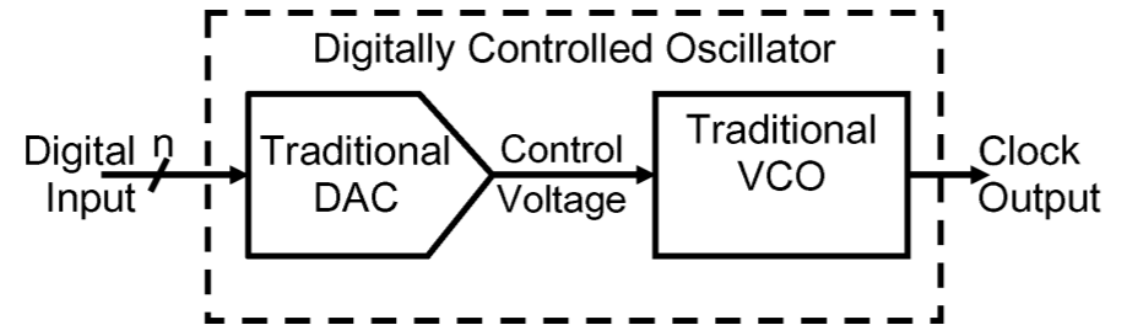


Digitally Controlled Oscillator (DCO)

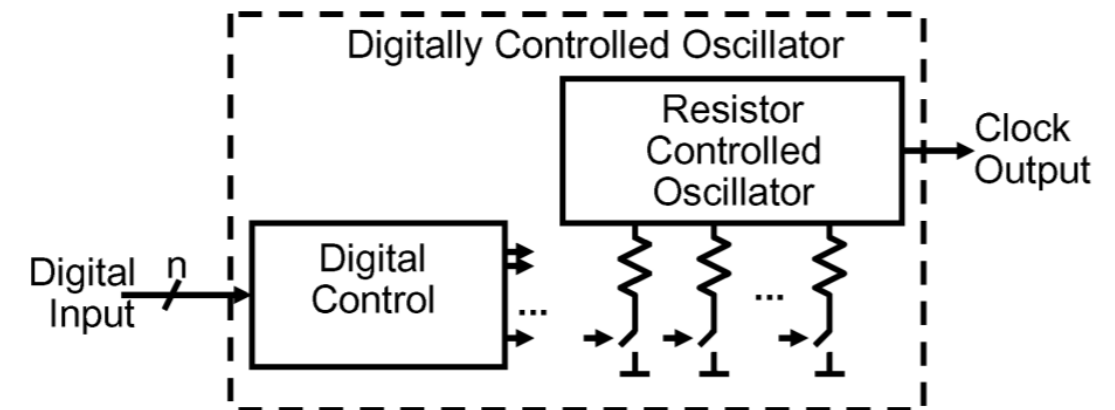


DCO in digital phase locked loop

- Input is digital control bits
- Output is oscillator frequency/phase
- DCO gain is measured in MHz/bit



DCO by DAC + VCO

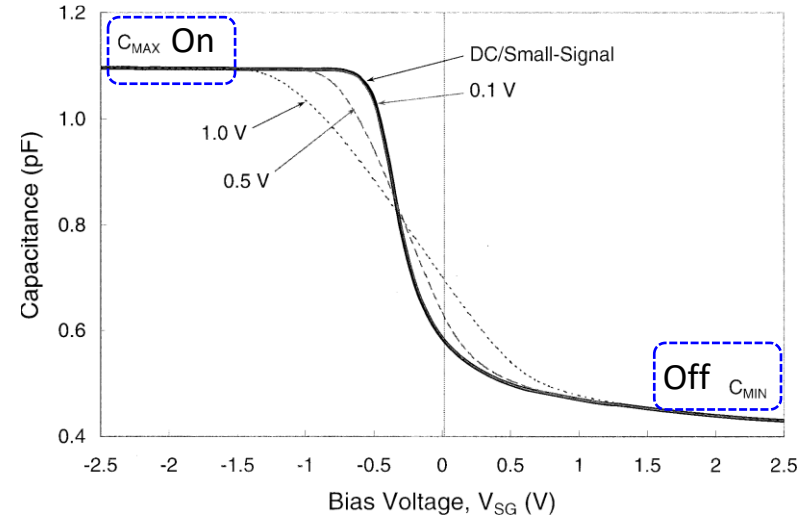
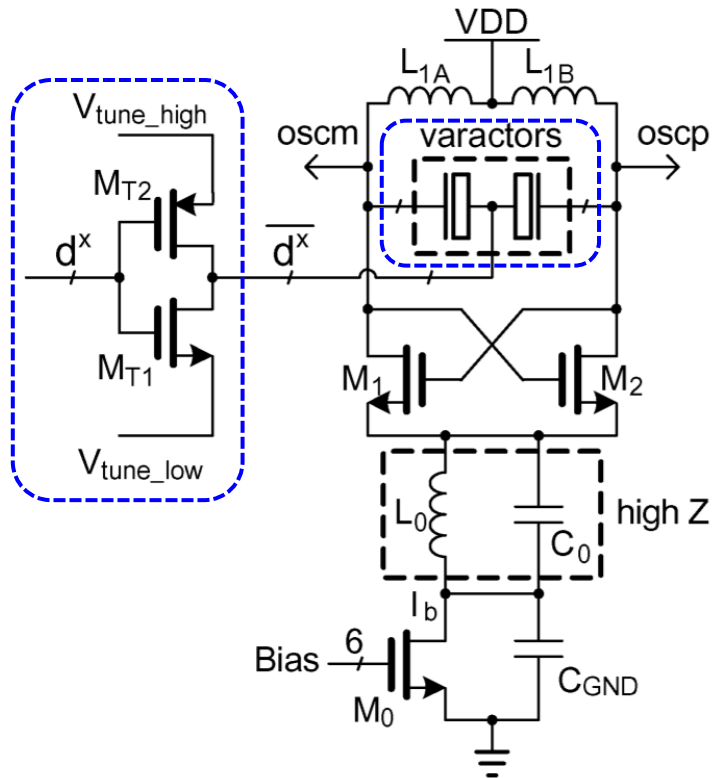


DCO by digitally controlling a physical oscillator parameter

Ref [2]



LC DCO with Varactor Tuning

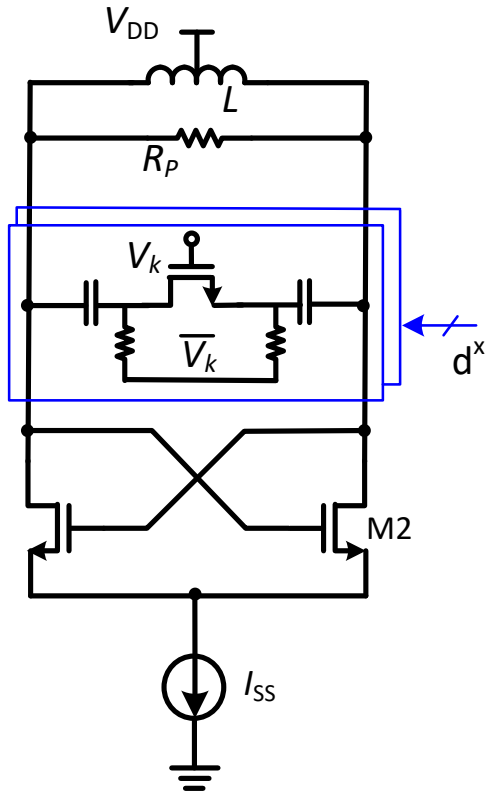


- Capacitor or varactor arrays controlled by digital bits
- Varactor
 - Switches between on and off states
 - DC coupled: minimum ΔC and frequency resolution limited by minimum varactor size
 - AC coupled to reduce minimum ΔC

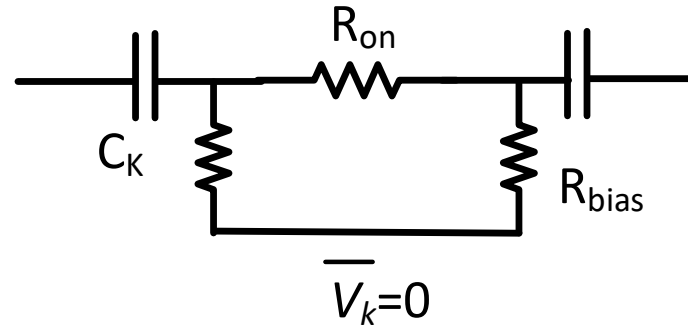
Ref [3, 4]



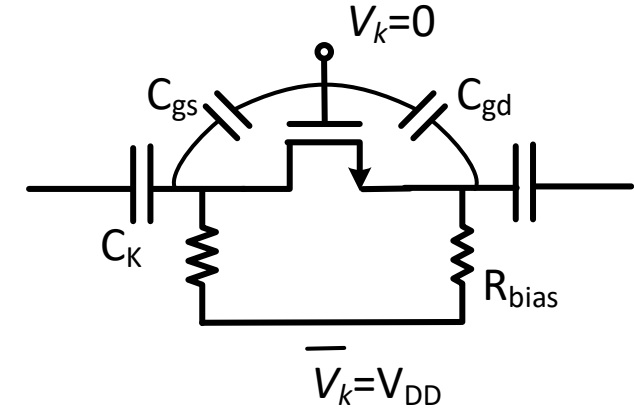
LC DCO with Switched Capacitor Tuning



$$V_k = V_{DD}, V_{GS} = V_{GD} = V_{DD}, \\ C_{on} = C_K/2, R_{on} \ll R_{bias}$$



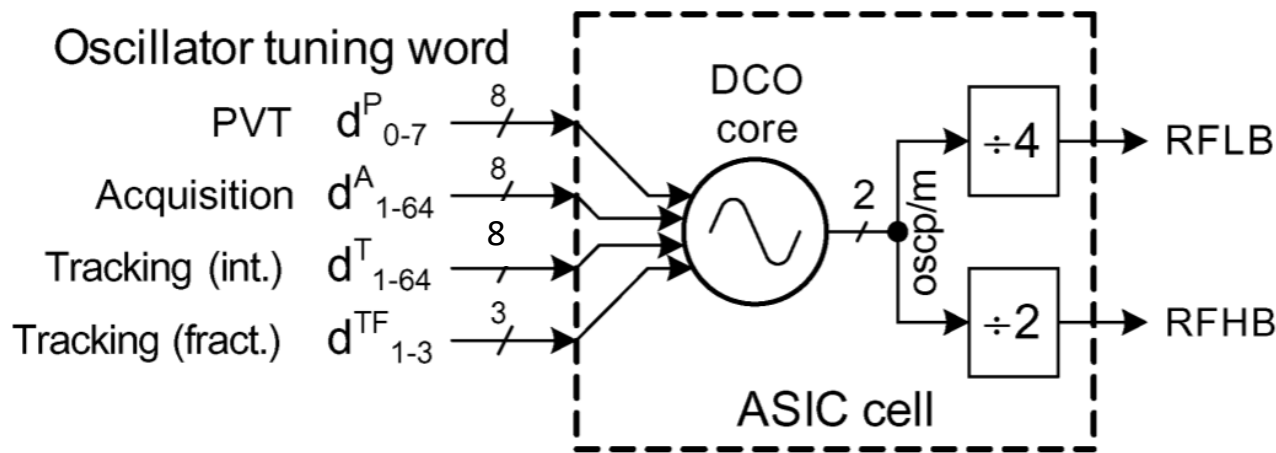
$$V_k = 0, V_{GS} = V_{GD} = -V_{DD}, \\ C_{off} = C_{gs} // C_{gd} // (C_K/2)$$



- Capacitor or varactor arrays controlled by digital bits
- Switched capacitor
 - Switches between on and off states
 - For small ΔC , C_{on} and C_{off} ratio becomes very small
 - Additional AC coupling cap to reduce ΔC
 - Large C_{off} to achieve the same ΔC -> frequency tuning range penalty



DCO Tuning Word



Example of oscillator tuning word

Varactor bank	Input	Weighting	Step size Δf
PVT	d^P	8-bit binary	$\Delta f^P = 4 \text{ MHz}$
Acquisition	d^A	64-bit unit	$\Delta f^A = 200 \text{ kHz}$
Tracking int.	d^T	64-bit unit	$\Delta f^T = 12 \text{ kHz}$
Tracking fract.	d^{TF}	3-bit unit	$\Delta f^T = 12 \text{ kHz}$

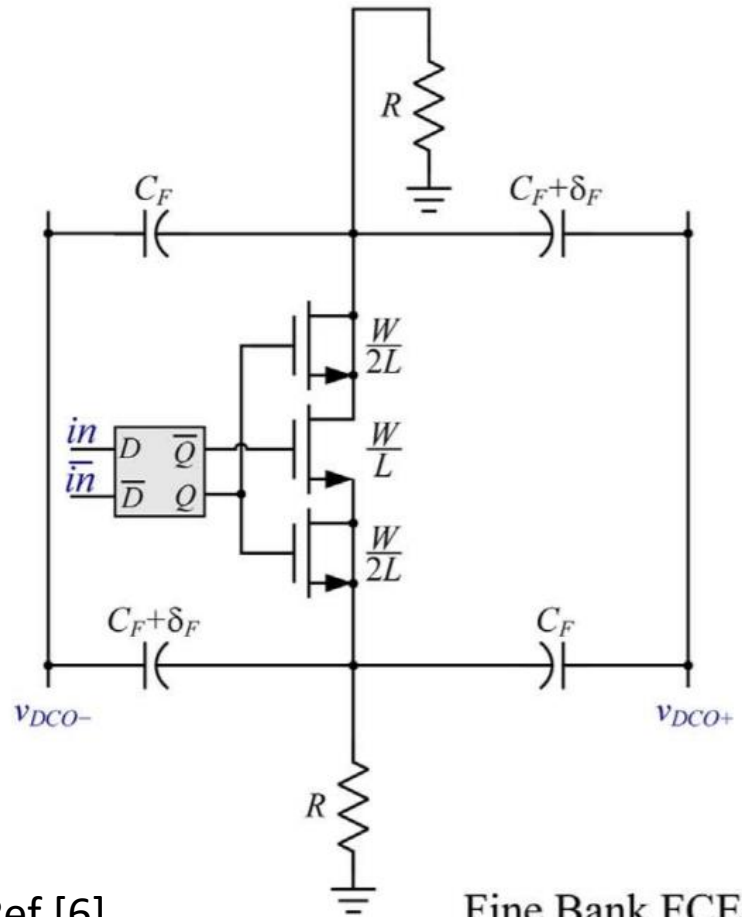
For cellular application, <100Hz resolution is needed

- Oscillator tuning word (generated from frequency control word FCW)
 - d^P : tune oscillator close to target frequency over PVT, binary weighted – coarse tune
 - d^A : tune oscillator further toward target frequency, thermometer coded – fine tune
 - d^T : track oscillator frequency, thermometer coded
 - d^{TF} : track oscillator frequency with delta-sigma modulator dithering, thermometer coded
 - Sequence
 - $d^P \rightarrow d^A \rightarrow d^T$ and d^{TF}
 - Coarse tune with d^P followed by fine tune with d^A during frequency locking; freeze both d^P and d^A
 - Frequency track with d^T and d^{TF} . d^{FT} bits generated by a DSM for quantization noise shaping and further improve resolution

Ref [3]



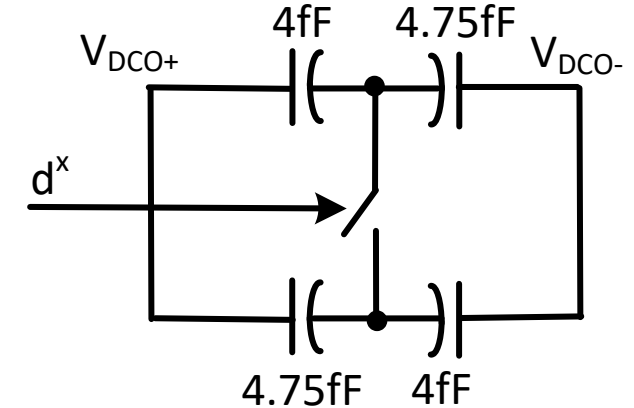
Switched Capacitor Tuning with High Resolution



$$\omega = \frac{1}{\sqrt{LC}}, \quad \left| \frac{d\omega}{\omega} \right| = \frac{1}{2} \left| \frac{dC}{C} \right|$$

2GHz carrier frequency, 100Hz resolution, tank cap 2pF
 $\rightarrow \Delta C \sim 2 \times 10^{-19} \text{F}, 0.2 \text{aF}$

- Switch on $C_{on} = \frac{1}{2} (2C_F + \delta_F)$
- Switch off $C_{off} = 2 \frac{C_F(C_F + \delta_F)}{2C_F + \delta_F}$
- Delta C: $\Delta C = C_{on} - C_{off} = \frac{\delta_F^2}{4C_F + 2\delta_F}$
- For $C_F = 4 \text{fF}$ and $\delta_F = 0.75 \text{fF}$, $\Delta C = 32 \text{aF}$



References

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